Analytics Capstone Project (ANCP)

# Sparta: Wardrobe Demand Analytics for Brands

Deliverable D4: Modelling checkpoint (baselines, candidate models, experiment log)

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|---|---|
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| Linked outcomes | CO3 |
| Files | ANCP_TM_D4_v1.pdf; code, experiment_log.csv, validation_design.json, environment.yml and README in submissions/D4 of the team repository, tag d4-v1 |

# Sparta: forecasting segment demand, baselines first

ANCP D4 · Modelling checkpoint · Group M · Linked outcome CO3 · Data: data-v1.0 (D2), H&M 10% customer sample, 950 weekly segments (131 Menswear), a stand-in for Sparta purchase and demand data while the pilot has none

## Bottom line

On the validation origins, the global LightGBM forecaster (M3) has a Menswear WAPE of **0.455** against **0.587** for the charter baseline (same week last year): **22.5% lower** (95% CI 17.2% to 27.7%, bootstrap over segments). This clears the charter target of 15% or more. Two cautions. First, the same-week-last-year baseline is weak here: a plain last-4-week mean (B1) already scores 0.500, and M3 beats it by only 9.0%. Second, the simple scaled seasonal naive (M1, 0.462) is statistically level with M3 (CI -3.1% to +6.3%). The 80% intervals cover 80.5% of Menswear actuals on the check origins (target 75% to 85%). **The test weeks are untouched** (charter risk R4); D5 scores M3 and M1 on them once.

## 1 Problem type and why these model families fit

The charter's primary decision is which menswear category-colour segments a brand should add or expand. Until the pilot has data, D4 builds the part the charter assigns to public data: a **demand forecast per segment** (Table 3, WAPE against same-period-last-year, with an 80% interval). We frame it as **weekly count forecasting for many related, short, seasonal series**, 1 to 4 weeks ahead. A segment is a product type x master colour within a department (H&M has no fit field). Sparta is menswear, so Menswear segments are the primary score; models learn from all departments, which share seasonality.

| Family | Why it fits (evidence) | Risk we watch |
|---|---|---|
| Seasonal naive and naive (B0, B1) | The charter names same-period-last-year as the baseline. The last-4-week mean is the standard naive forecast [1] and tests whether seasonality helps at all when demand turns over within weeks (D3 I5). | Two years of history: one past season per week. |
| Scaled seasonal naive (M1) | Last year's shape times this year's level. Explainable to a brand and robust on 2 years of data; three tuned knobs. | Breaks when the level shifts suddenly (2020). |
| Poisson GLM (M2) | Linear in log demand; counts with many small values; one model for all segments. | No interactions, e.g. season x product type. |
| Global gradient boosting, Tweedie loss (M3) | One model across 950 series pools seasonality; Tweedie suits zero-heavy counts. Global LightGBM models led the M5 retail forecasting competition [2][3]. | Overfitting validation; opaque to a brand. |

## 2 Validation design, declared before tuning

Fixed in **validation_design.json** on 2026-11-12 before the first run; its hash (ce0c8ff14330) is on every log row. A **rolling origin with an expanding window** [1][4]: at each origin the models see only earlier weeks, and global models are refitted on training origins whose targets are already observed. Origins span pre-COVID and 2020 weeks, because D3 (I9) warned against tuning on 2020 alone. Segments are kept if they average 2+ units a week before the first validation target (week 58), so eligibility cannot leak. 23,066 units in the first partial week are dropped; kept segments hold 97.0% of units.

| Part | Weeks (7-day, ending Tuesday) | Use |
|---|---|---|
| Train | From 2018-09-26, expanding up to each origin | Features and model fitting; last-year features from week 52 on |
| Validation | 9 origins every 4 weeks; targets 2019-11-20 to 2020-07-21 | All tuning and selection (39 runs) |
| Test | 2 origins; targets 2020-07-22 to 2020-09-22 | Declared, not run. Scored once in D5 (charter R4) |

## 3 Baselines first, then candidate models

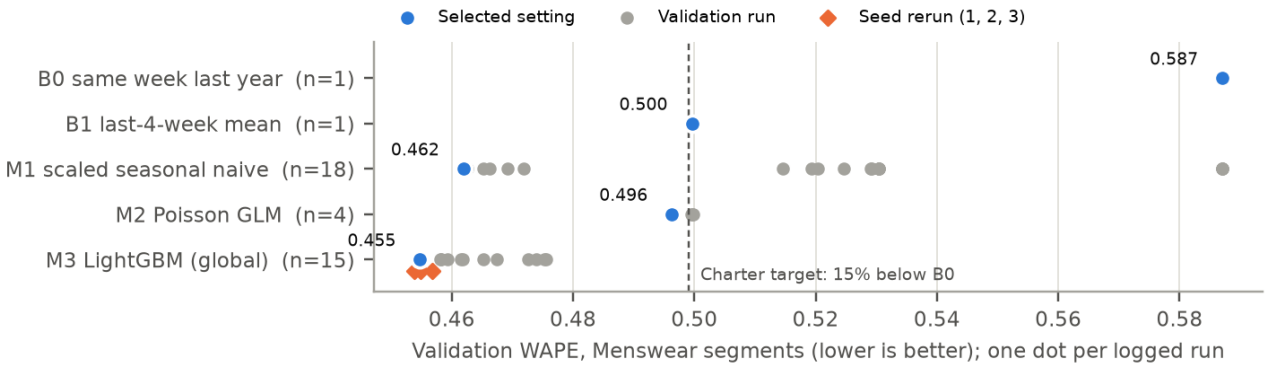



Figure 1. Every validation run. The dashed line is the charter target (15% below B0). Runs below it: B1 0 of 1, M1 6 of 18, M2 1 of 4, M3 15 of 15. The GLM clears it only narrowly.

| Model | Tuning approach | Runs | Selected setting | WAPE Menswear | WAPE all | Bias Menswear |
|---|---|---|---|---|---|---|
| B0 same week last year | None (charter baseline) | 1 | lag 52 weeks | 0.587 | 0.584 | +0.134 |
| B1 last-4-week mean | None (rule fixed in advance) | 1 | 4 weeks | 0.500 | 0.411 | -0.028 |

| Model | Tuning approach | Runs | Selected setting | WAPE Menswear | WAPE all | Bias Menswear |
|---|---|---|---|---|---|---|
| M1 scaled seasonal naive | Full grid 3x3x2: recent window {4, 8, 13} wk x shrink {0, 0.5, 1} x last-year smoothing {1, 3} wk | 18 | window 8, shrink 1.0, smoothing 3 | 0.462 | 0.450 | +0.002 |
| M2 Poisson GLM | Grid over L2 penalty alpha {0.001, 0.01, 0.1, 1}; refit at each origin | 4 | alpha 1.0 | 0.496 | 0.429 | +0.088 |
| M3 LightGBM (global) | Random search, 12 trials (seed 42) over leaves, learning rate, trees, min child, column and row sampling, Tweedie power; refit at each origin; best re-run with seeds 1-3 | 15 | num_leaves 31, learning_rate 0.0313, n_estimators 800, min_child_samples 50, colsample_bytree 0.8, subsample 0.7, tweedie_variance_power 1.1 | 0.455 | 0.389 | -0.005 |

## 4 Is the gain real?

| Comparison (validation, Menswear) | WAPE difference | 95% CI | Relative reduction (95% CI) | Reading |
|---|---|---|---|---|
| M3 LightGBM - B0 last year | -0.132 | -0.182 to -0.093 | 22.5% (17.2% to 27.7%) | Clear gain |
| M3 LightGBM - B1 naive | -0.045 | -0.071 to -0.024 | 9.0% (5.1% to 13.2%) | Clear gain |
| M3 LightGBM - M1 scaled | -0.007 | -0.032 to +0.013 | 1.6% (-3.1% to 6.3%) | No detectable difference |
| M3 LightGBM - M2 GLM | -0.042 | -0.066 to -0.022 | 8.4% (4.8% to 12.2%) | Clear gain |
| M1 scaled - B0 last year | -0.125 | -0.172 to -0.086 | 21.3% (15.2% to 26.7%) | Clear gain |

Paired bootstrap resampling whole Menswear segments (131 segments, 2,000 draws, seed 7), so the correlation across a segment's weeks is kept. **Seed variance:** the selected M3 setting scores 0.454 to 0.457 over seeds 42, 1, 2, 3, a range of 0.003, smaller than any gap read as clear above. **By origin** (Figure 2), B0 is worst at 2020-04-14 (0.74), M1 is worst at 2020-03-17 (0.57), M3 is worst at 2020-03-17 (0.59); the spring 2020 closures hurt every model, and last year's pattern fails longest. **By horizon**, the best model is M3 at 1 wk, M1 at 2 wk, M3 at 3 wk, M1 at 4 wk, which supports carrying M3 and M1 to D5.

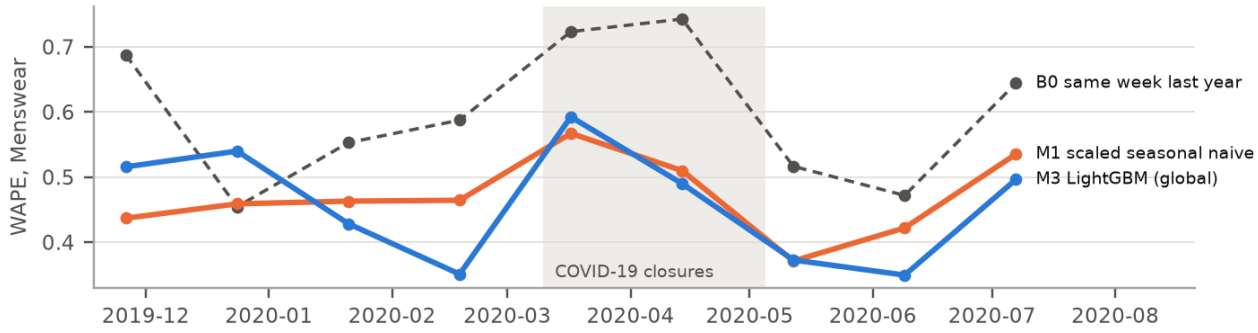



Figure 2. Menswear WAPE at each validation origin (date of the first forecast week).

## 5 Uncertainty: 80% prediction intervals

Split conformal intervals [5] on the log ratio (actual + 1) / (forecast + 1), one pair of quantiles per horizon, calibrated on origins 59 to 75 and checked on later origins 79 to 91. Coverage: **80.5%** Menswear, 79.6% all segments, inside the charter's 75% to 85% band. The intervals are wide (at 1 week the 80% band runs from about 0.44x to 1.79x the forecast), which is the honest size of week-level error on a 10% sample.

## 6 Experiment log and reproducibility

**experiment_log.csv** has one row per run (40 rows: 39 model runs and 1 interval check). Each row records the run id, UTC timestamp, stage, origins, model, data version (data-v1.0 plus the SHA-256 of the transactions file), feature list, parameters as JSON, random seed (n/a for deterministic rules and the GLM), Menswear WAPE, all-segment WAPE, bias, forecasts scored, run time, design hash and code hash. Top M3 features by gain: lag1 64%, m4 27%, ptype 2%, lag2 2%. **Independent check: validate_metrics.py** rebuilds the Menswear panel with separate code and matches the logged B0 and B1 scores (0.58706, 0.49959; match = True).

**Run it: conda env create -f environment.yml** (Python 3.13.5; pandas, scikit-learn, LightGBM pinned in requirements.txt). Put data-v1.0 at **../D2/data-v1.0** or pass **--data**; the loader stops if the checksum differs. Then run **python -I run_experiments.py** (about 2 minutes on a 16-thread laptop), **python -I validate_metrics.py** and **python -I build_report.py**. The data stays out of the repository (Kaggle terms, as in D2).

## 7 Limits and what D5 does

(1) The charter's first metric, precision@10 of gap segments, needs Sparta wishlist and ownership data that H&M cannot supply; it stays a D5 item on aggregated pilot figures only (BR-28: no model is trained on participant data). (2) H&M Menswear is European, 5.8% of units and has no fit field, so the scores show the method, not Indian menswear demand. (3) Only two years of history: the last-year features rest on one season. (4) D5 scores M3 and M1 once on the two test origins with the settings frozen in **outputs/selected_configs.json**, and reports both against B0 and B1.

## References

[1] Hyndman, R. J. & Athanasopoulos, G. (2021). Forecasting: Principles and Practice, 3rd ed., §5.2 simple methods, §5.10 time series cross-validation. OTexts. [2] Makridakis, S., Spiliotis, E. & Assimakopoulos, V. (2022). M5 accuracy competition: Results, findings, and conclusions. International Journal of Forecasting, 38(4). [3] Ke, G. et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. NeurIPS 30. [4] Kapoor, S. & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. Patterns, 4(9). [5] Angelopoulos, A. N. & Bates, S. (2021). A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv:2107.07511. [6] H&M Group (2022). H&M Personalized Fashion Recommendations [Data set]. Kaggle.

AI-tool disclosure: the team used Claude (Anthropic) to write the code and draft this report; every number is generated from the experiment log by build_report.py and was re-checked with separate code.